

ABRA GEIGER

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EDUCATION

Cornell University, Ithaca, NY

B.A., Physics and Math, Minor in Astronomy, 2022 to 2026 (expc.)
GPA: 4.0/4.3

PEER-REVIEWED PUBLICATIONS

1. *The NANOGrav 12.5-Year Data Set: Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1903+0327*
Geiger, A., et al. (36 authors), 2025
ApJ 986 191

OTHER PUBLICATIONS

1. *The Frequency-Dependent Scattering of Pulsar J1903+0327*
Geiger, A. & Lam, M. T., 2022
NANOGrav Memo Series No. 8, Tech. rep., NANOGrav

PUBLICATIONS IN PREPARATION

1. *Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1935+1616*
Geiger, A., et al.
To be submitted to *ApJ*
2. *Comment on “Pinpointing Gravitational Waves Via Astrometric Gravitational Wave Antennas”: Re-Evaluating The Potential for Gravitational Wave Detection with Stellar Binary Astrometry*
Geiger, A. & Flanagan, E.
To be submitted to *Nature*

RESEARCH EXPERIENCE

Professor James Cordes Astronomy Research Group

Research Assistant

Cornell University

9/2022 - Present

- Analyze interstellar scattering utilizing python for application to pulsar timing as a member of the North American Nanohertz Observatory for Gravitational Waves (NANOGrav)
- Write peer-reviewed scientific publications on scattering research and conduct literature reviews
- Collaborate and discuss weekly project reports with team of professors and researchers
- Co-authored accepted proposals to observe with the Green Bank telescope in West Virginia for 45 hours in 2024 and the Murchison Widefield Array in Australia for 3 hours in 2023 to study pulsar scattering

Professor Eanna Flanagan Theoretical Physics Research Group

Research Assistant

Cornell University

1/2025 - Present

- Study potential astrometric methods of gravitational wave detection and quantify their limits

- Quantify and correct a critical error found in a nature publication
- Review the literature and perform mathematical modeling for a senior honors thesis on gravitational wave memory

Professor Vikram Ravi Astronomy Research Group

California Institute of Technology

Research Assistant

6/2024 - 8/2024

- Simulated spectroscopic absorption and emission from circumgalactic clouds of gas and dust to better deduce circumgalactic cloud properties from optical observations
- Conducted optical spectroscopic observations of globular clusters at the Keck and Palomar observatories
- Prepared a final report to be developed into a publication focusing on forecasting for upcoming spectroscopic surveys of the Milky Way

Professor Michael Lam Astronomy Research Group

Rochester Institute of Technology

Research Assistant

9/2021 - 8/2022

- Measured and analyzed interstellar scattering variations for millisecond pulsar J1903+0327 utilizing python
- Modeled autoregressive variations in timing residuals for millisecond pulsar B1937+21 utilizing python
- Presented and discussed progress in weekly meetings with a team of pulsar timing researchers

AWARDS AND FELLOWSHIPS

Barry Goldwater Scholarship (\$7,500) 2025

Prestigious national scholarship awarded to undergraduates pursuing careers in research

Society of Physics Students Jocelyn Bell Burnell Outstanding Leadership Scholarship (\$6,000) 2025

Awarded to one undergraduate each year on the basis of academic performance and leadership in their department

FUTURE of Physics at Caltech ($\approx$ \$1000) 2025

Prestigious all-expenses paid conference for prospective graduate students

Cornell Astronomy Cranson W. and Edna B. Shelly Undergraduate Research Award (\$600) 2025

Awarded to one undergraduate in the Cornell Astronomy Department on the basis of research achievement

Cornell Quill and Dagger Senior Honors Society 2025

Distinction recognizing the top 1% of each Cornell graduating class with respect to leadership, character, and dedication to service

Caltech Summer Undergraduate Research Fellowship (\$7,740) 2024

Fellowship funding a 10-week, full-time summer research experience at Caltech

Ronald E. McNair Scholar ($\approx$ \$15,000) 2023

National program funding attendance at conferences, graduate school visits, an advocacy trip to Capitol Hill, and a 9-week, full-time summer 2025 research experience at Cornell, as well as providing guidance in pursuit of a PhD

Cornell Arts & Sciences Nexus Scholar (\$7,000) 2023

Fellowship funding a 9-week, full-time summer research experience at Cornell

PRESENTATIONS

Cornell Office of Inclusive Excellence Summer Research Symposium 8/2025

Talk and Poster: *Gravitational Waves, Memory, and Astrometry*

Cornell Undergraduate Physics Poster Session 12/2024

Poster: *Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1903+0327*

NANOGrav Fall Collaboration Meeting	10/2024
Talk: <i>Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1903+0327</i>	
Caltech Student-Faculty Programs Final Presentations	8/2024
Talk: <i>Tracing the Milky Way Circum-Galactic Medium With Optical Spectroscopy</i>	
Cornell Astronomy Undergraduate Research Poster Forum	5/2024
Poster: <i>Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1903+0327</i>	
• 2024 Friends of Astronomy Undergraduate Poster Winner (\$150)	
Ronald E. McNair Scholars & Undergraduate Research Conference	3/2024
Talk: <i>Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1903+0327</i>	
American Astronomical Society Winter Meeting	1/2024
Poster: <i>Probing Interstellar Turbulence and Precision Pulsar Timing with PSR J1903+0327</i>	
Cornell Arts & Sciences Nexus Capstone Presentations	7/2023
Talk: <i>Probing Interstellar Turbulence and Its Intermittency Through Observation of a Millisecond Pulsar</i>	
Presentation to Fairport High School Physics Club	6/2023
Talk: <i>Detecting Gravitational Waves with Pulsars</i>	
Cornell Astronomy Undergraduate Research Poster Forum	5/2023
Poster: <i>The Frequency-Dependent Scattering of Pulsar J1903+0327</i>	
Cornell Undergraduate Research Board Spring Symposium	5/2023
Poster: <i>The Frequency-Dependent Scattering of Pulsar J1903+0327</i>	
• Best in Category for Physical Sciences Winner	
APS Conference for Undergraduate Women in Physics	1/2023
Poster: <i>The Frequency-Dependent Scattering of Pulsar J1903+0327</i>	
Rochester Institute of Technology Undergraduate Research Symposium	7/2022
Talk: <i>The Frequency-Dependent Scattering of NANOGrav Pulsar J1903+0327</i>	

TEACHING EXPERIENCE

Physics Learning Strategies Center	Cornell University
<i>Tutor</i>	1/2025 - Present
• Tutor students from a wide variety of physics courses, including mechanics, electricity and magnetism, quantum mechanics, and more for about 5 hours per week	
PHYS 1116: Mechanics and Special Relativity	Cornell University
<i>Undergraduate Teaching Assistant</i>	1/2023 - 12/2023
• Collaborate with professors and graduate student TAs to support students in augmented introductory physics • Host weekly office hours to aid students in homework completion and content comprehension	

SCIENCE LEADERSHIP AND SERVICE

Cornell Society of Physics Students

Cornell University

President

1/2025 - Present

- Facilitate the organization of physics speaker events, career panels, social events, mentorship programs, local outreach, the Cornell Physics Wiki, an annual poster session, and an annual formal
- Host about a dozen events per semester
- Serve on the CEDI committee, and continue to facilitate DEI initiatives in the Cornell Physics Department (see more below)
- Organize the initiatives of the organization, promoting professional development and community

Vice President

12/2023 - 1/2025

- Organize and oversee ~ 10 physics events per semester, collaborating with the president and executive board
- Lead our undergraduate peer mentorship program and our undergraduate-graduate mentorship program
- Serve on the CEDI committee and complete the “[A Physicist Is](#)” project for the first cohort (more information below)

Women in Physics Chair

2/2023 - 12/2023

- Spearhead tri-weekly outreach for ~ 40 local fifth grade students by gathering and training volunteers, communicating weekly physics concepts, and coordinating with local fifth grade teacher and The Ithaca Physics Bus
- Plan and host ~ 10 physics events per semester: including professional development workshops, speakers, etc
- Serve as a student-faculty liaison on the Cornell Physics Committee for Climate, Equity, Diversity, and Inclusion (CEDI)
- Develop “[A Physicist Is](#)” under Prof. Holmes, a project to showcase the diversity of careers and identities in physics through informational graphics posted around the physical sciences complex

Cornell Astronomical Society

Cornell University

Keyholder

2/2023 - Present

- Manage sky tours and telescope observing for weekly open houses at Cornell’s Fuertes Observatory
- Write astrophysics columns in our newsletters geared towards the general public

The Stemminist Movement

Cornell Chapter

Outreach Co-Director

11/2022 - 5/2024

- Co-lead a committee of 5 members to organize ~ 10 STEM outreach programs for middle school students each year
- Won and executed a grant for \$2500 for a project to distribute 300 STEM kits to youth in the local area
- Train co-director and committee members in organizational workflow and event planning
- Foster an empowering community for women in STEM at Cornell and beyond through professional development workshops, invited talks, socials, and more

Rochester Museum and Science Center

Volunteer

Rochester, NY

12/2022 - 1/2023

- Present the “Science On A Sphere” exhibit and its connection to scientific research and discovery
- Monitor and tidy the museum and assist with other operational tasks as needed

OTHER LEADERSHIP AND SERVICE

Ballet & Books

Dance Teacher

Ithaca Chapter

9/2022 - Present

- Lead weekly ballet classes for ~ 15 students aged 6-9 years, managing a team of 2 teaching assistants

Treasurer

1/2023 - 12/2024

- Manage club logistics including semesterly show planning, fundraisers, finances, and funding allocation
- Won and executed a grant for \$2500 for our students dancewear and performance

Cornell Einhorn Center for Community Engagement

Pre-Orientation Service Trips Leader

Cornell University

1/2025 - Present

- Plan and facilitate a 4-day sleepover pre-orientation trip for new students at Cornell as one of 9 leaders
- Serve the local community via 3 full-day service trips
- Foster an inclusive, empowering, and lasting community among the new student participants

Cornell Tatkon Center for New Students

Orientation Leader

Cornell University

8/2023

- Facilitate small group sessions and large-scale events with new students
- Support and guide new students through the transition to campus life
- Engage in training for developing inclusive spaces, building connections, and strengthening teamwork

Court-Kay-Bauer Residence Hall Student Council

Vice President

Cornell University

9/2022 - 6/2023

- Plan and run events for the 275+ residents of Court-Kay-Bauer Hall every 1-to-2 months
- Collaborate with other Hall Councils and Residential Student Congress for larger scale events

SKILLS

Programming: Python, Java, Latex, HTML

Computer: UNIX, Microsoft (Word, Powerpoint, Excel, Office), Google Workspace

Language: Native Fluency in English, Intermediate French

AFFILIATIONS

Associate Member, North American Nanohertz Observatory for Gravitational Waves

Member, American Astronomical Society

Member, American Physical Society

Member, Society of Physics Students